Beyond the Known: Challenging Beal's Conjecture with Fresh Eyes

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Preface

And just as a disclaimer, the try is completely innovative and bold, so please before you start judging please change your lenses ;). Thanks in advance my dears!
Also note: I AM NOT A PROFESSIONAL MATHEMATCICIAN AND THAT THIS IS ONLY AMATEUR TEENAGE WORK DONE WITH VERY LITTLE RESSOURCES, anyway sending LOVE!

Abstract

This is in some kind a simple and innocent try to tackle one of the craziest conjectures by a random kid out in this world, only with a pen, papers – a lot –, an old Nokia phone – RIP – for instant research (it had data, don't worry;)) and let's not forget the help to grasp some advanced math concepts from a guy in my bus; thank you;). YEAH! Life's easy, anyway, here you go.

Introduction

The Beal Conjecture states that for any natural numbers other than 0,

$$a^x + b^y = c^z,$$

with $a, b, c > 2$, only exists if and only if a, b and c have a common factor other than 1. Note that all numbers here are in $\mathbb{N}^*$.

Try

Let

$$x = \alpha + 2, \quad y = \beta + 2, \quad z = \gamma + 2$$

with $\alpha, \beta, \gamma \in \mathbb{N}^*$.

The expression $a^x + b^y = c^z$ becomes

$$a^{\alpha+2} + b^{\beta+2} = c^{\gamma+2}$$

which is just

$$a^\alpha \cdot a^2 + b^\beta \cdot b^2 = c^\gamma \cdot c^2.$$

And let

$$m = a^\alpha, \quad n = b^\beta, \quad k = c^\gamma,$$

then the expression just becomes

$$m \cdot a^2 + n \cdot b^2 = k \cdot c^2.$$

And we know that original Pythagorean triplets come without a common factor, while all others are just original ones with a certain factor; example: if (a, b, c) is a Pythagorean triplet then (ha, hb, hc) is also a triplet.

Proof by Contradiction

Then if $(m \cdot a^2, n \cdot b^2, k \cdot c^2)$ is an original triplet, in other words:

$$\gcd(m \cdot a^2, n \cdot b^2, k \cdot c^2) = 1,$$

A, B and C should exist in $\mathbb{N}^*$ as

$$m \cdot a^2 = A^2, \quad n \cdot b^2 = B^2, \quad k \cdot c^2 = C^2,$$

from which the statement

$$(a\sqrt{m}, b\sqrt{n}, c\sqrt{k}) \in \mathbb{N}^*$$

should be true.

AND SINCE WE ARE TRYING TO PROVE BEAL'S CONJECTURE AND NOT DISPROVE IT THEN WE SHOULD JUST PROVE:

$$a^x + b^y = c^z$$

can't exist in $\mathbb{N}^*$ when x, y and z are all even.

Conclusion

In order to prove or disprove – which is unlikely – Beal's Conjecture you have to prove or disprove my new conjecture that I declare: **Amine's Pythagorean Triplets Conjecture**.

The Amine's Pythagorean Triplets Conjecture states that:

> *"NO PYTHAGOREAN TRIPLETS THAT ARE ALL PERFECT POWERS THEMSELVES EXIST."*

The proof is left as an exercise for the reader, if you do it contact me in this life or in the afterlife.

END